

H.I.T. Assistant for Baseball

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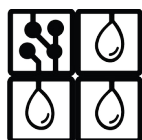


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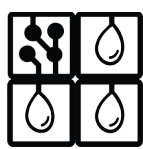
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Introduction

The goal of this project is to create an AI powered player trading assistant called Hyper-Intelligent Trading (H.I.T.). It uses historical player data to project a players future performance, determine a fair market value for a trade, and evaluate a players injury history.

Core Objectives

This section will cover the objectives that this project is trying to achieve.

Player Performance Projections

H.I.T. uses player performance historical data to try and project a players performance. By using both standard statistics (hits, home runs, ERA) and advanced statistics like WAR (Wins Above Replacement), FIP (Fielding Independent Pitching), wOBA (Weighted On-Base Average), and SIERA (Skill-Interactive ERA) allows both modern and traditional baseball analysts to evaluate players. For players that do not have advanced statistics, majority can be calculates by the standard statistics.

Fair Market Value

A secondary objective for H.I.T. is evaluating a players fair market value. By comparing players with comparable performances and knowing public contract terms allows the system to evaluate a fair market value. Allowing general managers to evaluate a players addition to a team on both an financial and player performance perspective.

Injury Assessment

Frequent injuries can change a player's value. By providing a summary of a player's injury history and assigns a low, moderate, or high risk level based on past trends.

Challenges

Defensive Statistics outside of scope. Historical defensive data is inconsistent (UZR, Total Zone tracked at different times). H.I.T. focuses only on batting and pitching.

Advanced Stats Must Be Context-Aware. Some advanced metrics require park factors (e.g., wRC+, OPS+, ERA-), which lower-level leagues don't track.

Personal Goal

I will keep my personal goal vague in this proposal since I still want to receive concrete feedback on it.

My draft goal: Write a program that uses machine learning that is easy to integrate into other systems/projects.

